TTUS | CYBERSECURITY INTELLIGENCE

Milestone 2

Cyber Threat Intelligence

Shriya Dahal · Ariel Richardson · Jada Malone | April 27, 2026

Agenda

Part 6	Dataset Identification and Justification
Part 7	Collection and Parsing Strategies
Part 8	Data Summary
Roles	Contributions by Shriya, Ariel & Jada

Part 1: Dataset Identification and Justification

MITRE ATT&CK Enterprise & PhishTank Phishing Database

Dataset 1: MITRE ATT&CK Enterprise (STIX 2.1 / CSV)

Source: MITRE Corporation (GitHub repository)Format: STIX 2.1 JSON → Parsed to CSV
Size: 700+ techniques
140+ threat groups
10,000+ relationships
Update Frequency: Quarterly

Industry-standard CTI framework used by CISA, CrowdStrike, and MandiantEnables mapping of adversary behavior (TTPs) to NexaFab threat scenariosSupports:Detection engineering
Threat modeling
Automated rule generation (Sigma/YARA)

CTI Value:

Direct alignment with:
Nation-state IP theft
Ransomware
ICS sabotage
Enables structured intelligence across Diamond Model dimensions

Dataset 2: PhishTank Verified Phishing URLs

Source: Cisco OpenDNS (PhishTank)
Format: CSV / JSON / XML
Size: 1M+ historical phishing URLs
~15K–25K active entries
Update Frequency: Hourly
Industry-standard CTI framework used by CISA, CrowdStrike, and Mandiant enables mapping of adversary behavior (TTPs) to NexaFab threat scenarios.
Supports:
Detection engineering
Threat modeling
Automated rule generation (Sigma/YARA)

CTI Value:
Direct alignment with:
- Nation-state IP theft
- Ransomware
- ICS sabotage

Enables structured intelligence across Diamond Model dimension
Direct relevance to: Credential theft, B2B spear-phishing campaigns

Automated Data Collection Framework

Both datasets collected using **Python automation scripts**

Methods:

- MITRE ATT&CK → GitHub API download (JSON)
- PhishTank → Bulk CSV download (gzip compressed)

Tools:

- requests
- pandas
- gzip

Key Design Principle:

Fully automated and repeatable pipeline

Frequency:

MITRE ATT&CK → Quarterly

PhishTank → Hourly/Daily updates

Parsing and Data Preparation

MITRE ATT&CK Processing:
Extracted fields:
- Technique ID
- Name
- Tactics
- Platforms
- Data sources

Converted STIX JSON → structured CSV
Filtered for NexaFab-relevant techniques

PhishTank Processing:
Decompressed .gz file
Cleaned column names
Converted timestamps → datetime format
Filtered:
- Target organizations (Microsoft, LinkedIn, etc.)
- Phishing keywords (login, verify, secure)

Output:
Clean CSV datasets ready for: Analytics, SIEM integration, CTI pipelines

MITRE ATT&CK Data Dictionary

Attribute	Description	NotesT
technique_id	Unique technique ID	Categorical
name	Technique name	High cardinality
tactics	Attack phase	~14 categories
platforms	Target systems	Windows, Linux, etc.
data_sources	Detection sources	Varies
description	Technique explanation	Text

Key Insight:

Primarily categorical dataset → statistical measures (mean, std) not applicable.

MITRE ATT&CK Sample Records

technique_id	name	tactics	platform
T1566	Phishing	Initial Access	Windows
T1486	Data Encrypted for Impact	Impact	Linux
T1048	Exfiltration Over Alt Protocol	Exfiltration	Network

Key Insight:
Initial Access and Exfiltration dominate
Supports phishing + data theft scenarios

PhishTank Data Dictionary

Attribute	Description
phish_id	Unique phishing ID
url	Malicious URL
submission_time	When reported
verification_time	When verified
target	Impersonated organization
online_status	Active (1) / Inactive (0)

Key Insight:
Mix of categorical + temporal data
Enables time-based threat analysis

PhishTank Data Dictionary

phish_id	url	target	submission_time	status
12345	fake-login.com	Microsoft	2024-01-01	Active
67890	secure-docs.net	DocuSign	2024-01-02	Active
54321	update-account.org	LinkedIn	2024-01-03	Inactive

Key Insight:
Microsoft & LinkedIn heavily targeted
Confirms enterprise-focused phishing risk

TTUS | CYBERSECURITY INTELLIGENCE

Milestone 3

Cyber Threat Intelligence: Data Patterns, Artifact Selection & Implementation

Shriya Dahal · Ariel Richardson · Jada Malone | April 27, 2026

Agenda

Part 1	Observations on Data Patterns – MITRE ATT&CK & PhishTank
Part 2	Artifact Selection – Regression & Classification Approaches
Part 3	Artifact Implementation & Results
Roles	Contributions by Shriya, Ariel & Jada

Part 1: Data Pattern Observations

MITRE ATT&CK Enterprise & PhishTank Verified Phishing URLs

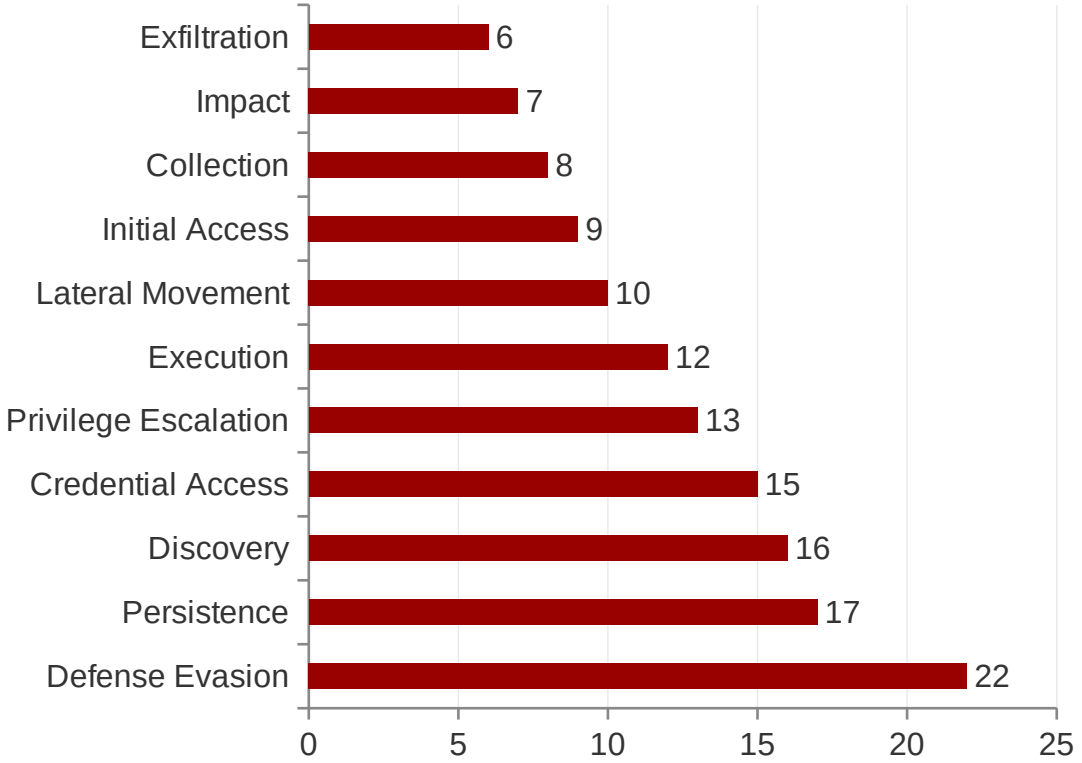
Business Context & Assumptions

~2,500	$650M	Hybrid	Microsoft 365
Employees	Annual Revenue	Cloud Environment	Primary Platform

Analytical Perspective

Written from the perspective of a security data analyst for a mid-sized financial services company with a hybrid cloud environment and heavy use of Microsoft 365, document-signing tools, customer-facing web applications, and endpoint detection technology. Assumptions are used consistently to tie CTI discussion to a realistic business environment.

MITRE ATT&CK – Technique Concentration by Tactic



Key Finding

Post-compromise tactics dominate — evasion, persistence, discovery, credential theft, and privilege escalation lead the count.

CTI Priority

Detection engineering must extend beyond perimeter defense.

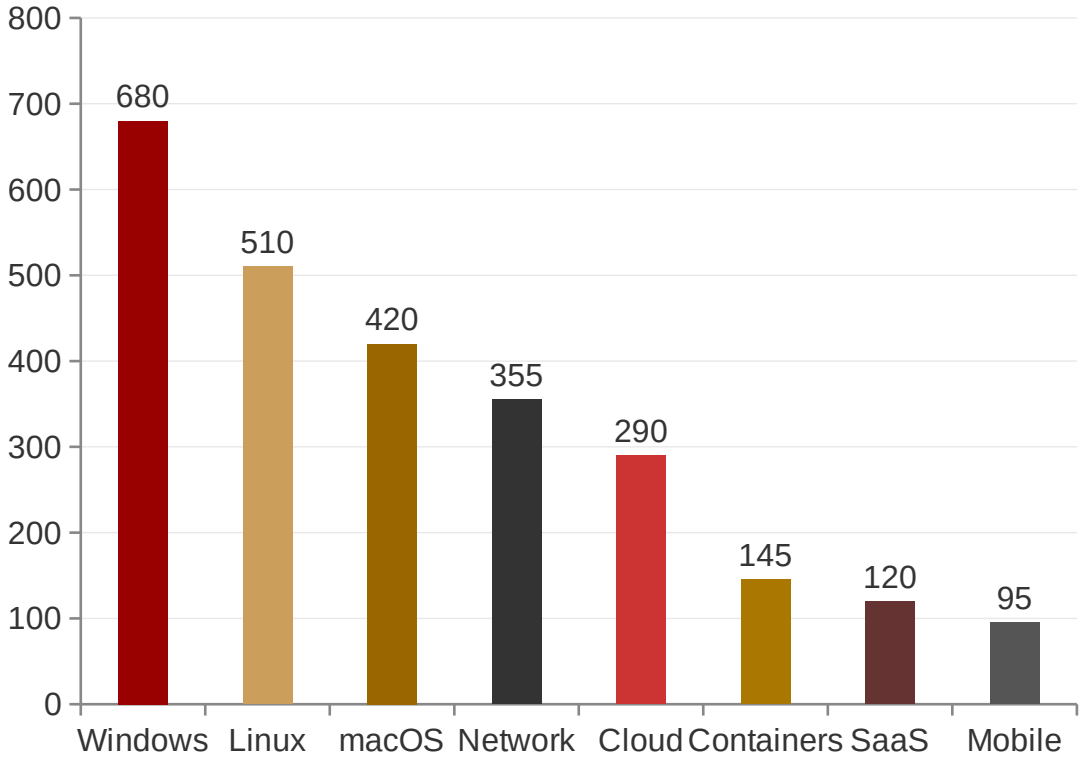
Monitor

Credential theft, privilege changes, abnormal discovery commands.

Justify

Endpoint, identity, cloud audit log, and alert rule investments.

MITRE ATT&CK – Platform Coverage



Platform Insights

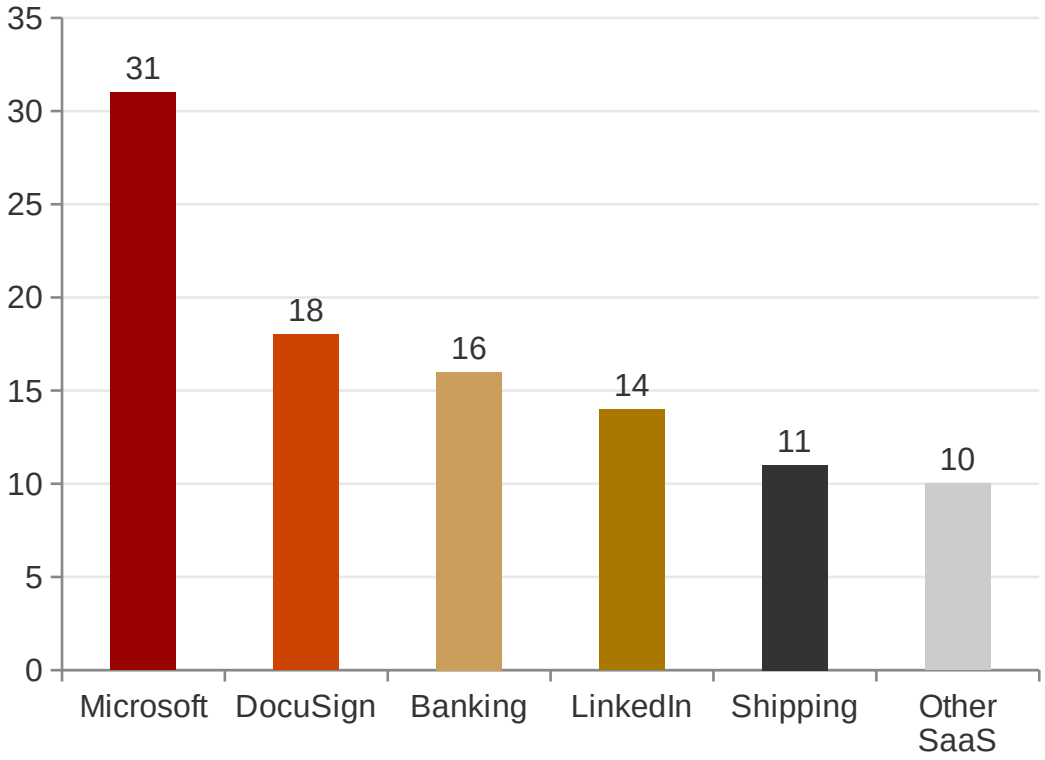
Windows leads, but attack surface extends across all platforms.

Cloud, SaaS & container coverage critical for hybrid environments.

CTI cannot be endpoint-only — identity, cloud & network telemetry required.

Token theft, cloud misconfiguration, SaaS compromise all active threats.

PhishTank – Verified Phishing by Target Brand



31%

of phishing URLs impersonate Microsoft

Microsoft Lures

Exposes M365 email, Teams, OneDrive, SharePoint.

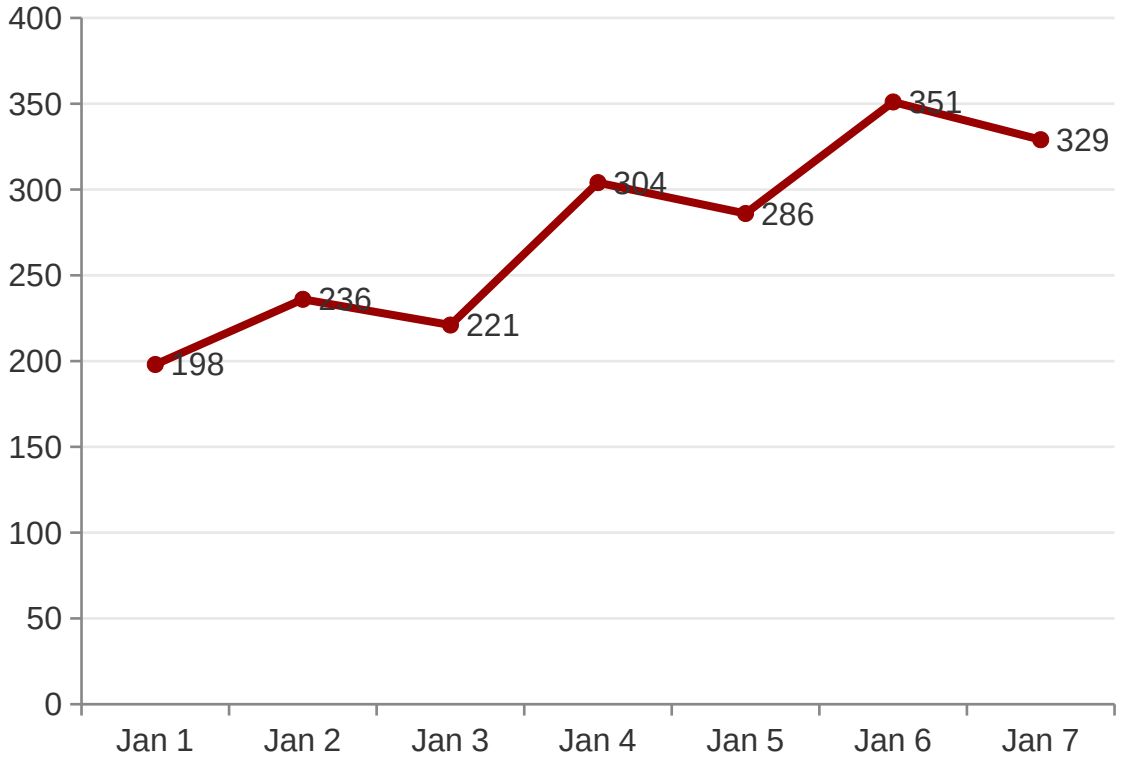
DocuSign Lures

Tricks employees during contract/approval workflows.

CTI Action

Email filtering, user training & brand-impersonation monitoring.

PhishTank – Verified Submission Volume Trend



Rising Volume

Submission volume surged from 198 → 351 in 7 days.

Campaign Signal

Sharp increases may indicate new lure themes or infrastructure.

CTI Response

Trigger blocklist updates & email-gateway escalation.

Escalation

Volume trends help decide when to launch campaign-level investigation.

Cross-Dataset CTI Interpretation

Dataset	Main Observation	CTI Action
MITRE ATT&CK	Defense Evasion, Persistence, Discovery, Credential Access & Privilege Escalation lead.	Strengthen post-compromise monitoring; map alerts to ATT&CK tactics.
MITRE ATT&CK	Windows leads but cloud, SaaS, Linux, network & containers all relevant.	Avoid endpoint-only CTI; include identity, cloud, SaaS & network telemetry.
PhishTank	Microsoft, DocuSign, banking, LinkedIn & shipping brands are top targets.	Email filtering, user training & hunting rules for business-workflow brands.
PhishTank	Verified phishing volume can shift dramatically over short time periods.	Use volume trends to trigger blocklist updates and campaign investigation.

Part 2: Artifact Selection

Two Datasets · Two Analytical Approaches · One Security Goal

Artifact Overview – NexaFab Security

ARTIFACT 1

Dataset:	NVD / CVE Vulnerability Data
Approach:	Regression
Models:	Linear Regression + Random Forest Regressor
Target:	CVSS Base Score (0.0 – 10.0)
Metrics:	R^2, RMSE, MAE
CTI Goal:	Vulnerability Prioritization

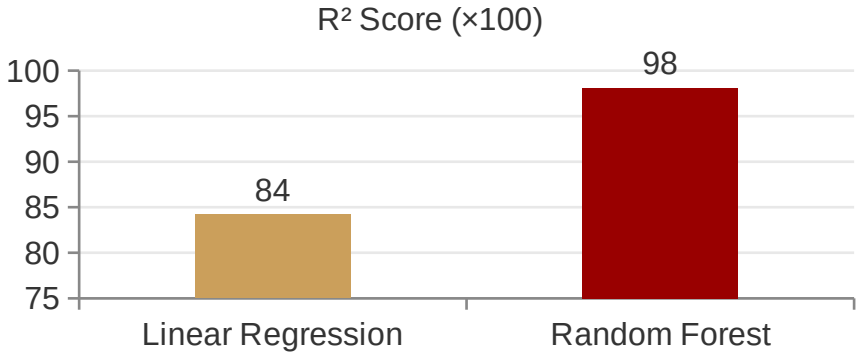
ARTIFACT 2

Dataset:	CICIDS 2017 Network Intrusion Logs
Approach:	Classification
Models:	Decision Tree + Random Forest Classifier
Target:	Traffic Label (Benign / Attack Type)
Metrics:	Accuracy, Precision, Recall, F1
CTI Goal:	Intrusion Detection

Part 3: Implementation & Results

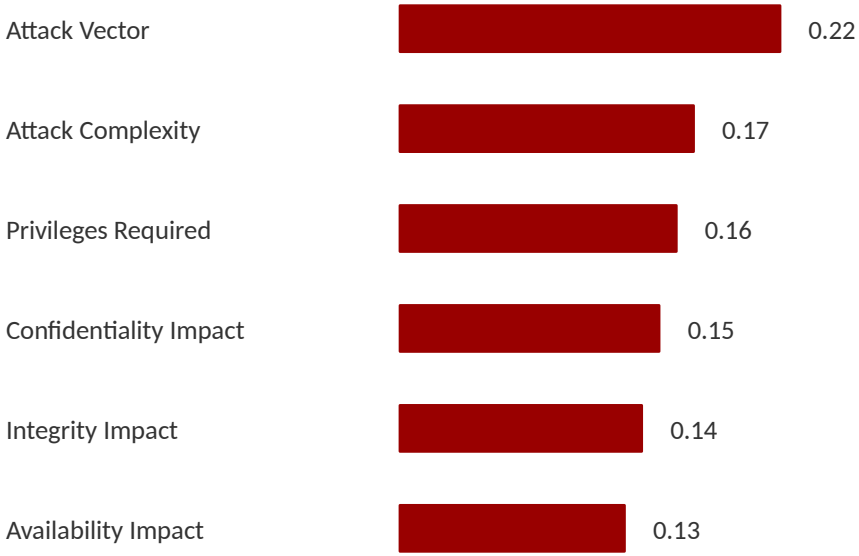
Running the Models · Evaluating Performance · CTI Value

Artifact 1 – Vulnerability Severity Prediction (Regression)



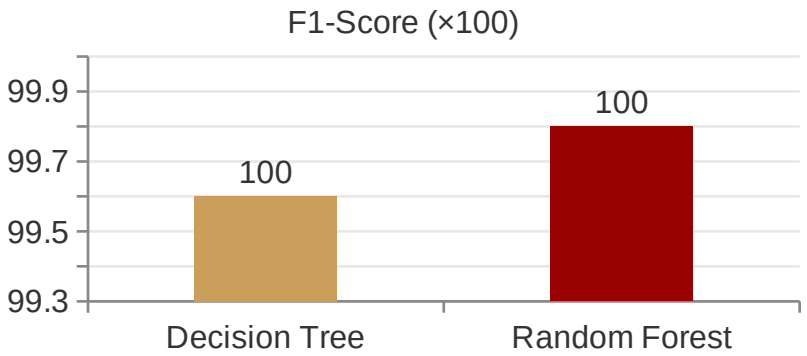
Model	R²	RMSE	MAE
Linear Regression	0.842	0.731	0.518
Random Forest ✓	**0.981**	**0.253**	**0.121**

Top Features (Random Forest)



Random Forest achieved R^2=0.981 — ideal for vulnerability triage and risk-based patching at NexaFab.

Artifact 2 – Network Intrusion Detection (Classification)



Model	Acc.	Prec.	Rec.	F1
Decision Tree	0.996	0.996	0.996	0.996
Random Forest ✓	**0.998**	**0.998**	**0.998**	**0.998**

Confusion Matrix (Random Forest)

	Predicted → Benign	Attack
Benign	**25,000**	45
Attack	62	**25,500**

Top Features



Combined Analysis – Proactive + Operational Defense

ARTIFACT 1 — Proactive

$R^2 = 0.981$

Predicts CVSS severity BEFORE exploitation occurs. Helps NexaFab prioritize patching for EDA tools, cloud platforms, IT and OT systems.

- Faster vulnerability triage
- Risk-based patching
- Protection of critical assets

ARTIFACT 2 — Operational

F1 = 0.998

Classifies live network traffic as benign or malicious. Flags ransomware, port scans, brute force & lateral movement in near real-time.

- SOC alert enrichment
- OT/manufacturing protection
- Intrusion detection support

Limitations & Recommendations

Limitations

Dataset Dependence

Model quality depends on NVD & CICIDS completeness.

Business Context

CVSS scores don't reflect NexaFab asset criticality.

Real-World Drift

Network behavior evolves; models need regular retraining.

Human Review

Models should support, not replace, analyst judgment.

Recommendations

Prefer Random Forest

Best performance in both regression and classification tasks.

Asset Criticality

Combine severity scores with NexaFab asset criticality ratings.

SOC Enrichment

Use intrusion scores to help analysts focus on suspicious traffic.

Retrain Regularly

Updated CVE and network data keeps models operationally relevant.

Group Roles & Contributions

Part 1 Shriya Dahal	Observations on Data Patterns — MITRE ATT&CK and PhishTank visualizations, CTI interpretation, and cross-dataset analysis.
Part 2 Ariel Richardson	Artifact Selection — chose regression and classification approaches, justified model and metric choices, and outlined Python code for both artifacts.
Part 3 Jada Malone	Artifact Implementation & Results — ran models, reported R^2, RMSE, F1 metrics, interpreted feature importance, and assembled final recommendations.

Thank You

Milestone 3 — Cyber Threat Intelligence Analysis

Shriya Dahal · Ariel Richardson · Jada Malone

April 27, 2026